

Assessing the Immune Cell Subset and Genetic Mutations in Patients With Palindromic Rheumatism Seronegative for Rheumatoid Factor and Anti-Cyclic Citrullinated Peptide

Chin-An Yang,¹ Ju-Pi Li,² Yi-Hua Lai,³ Ya-Ling Huang,⁴ Chien-Yu Lin,⁴ and Joung-Liang Lan³ 

Objective. The etiology underlying cases of palindromic rheumatism (PR) not associated with other rheumatic diseases in patients who are seronegative for rheumatoid factor and anti-cyclic citrullinated peptide (seronegative PR) is unclear. We aimed to investigate the immune cells and genes involved.

Methods. This was a single-center comparative study of 48 patients with seronegative PR and 48 healthy controls. Mass cytometry and RNA sequencing were used to identify distinct immune cell subsets in blood. Among the 48 seronegative PR patients, plasma samples from 40 patients were evaluated by enzyme-linked immunosorbent assay for cytokine levels, and peripheral blood samples from 25 patients were evaluated by flow cytometry for mononuclear cell subsets. Plasma samples from 21 patients were evaluated by real-time polymerase chain reaction for differential gene and protein expression, and samples from 3 patients were analyzed with whole-exome sequencing for gene mutations.

Results. Immunophenotyping revealed a markedly increased frequency of CD14+CD11b+CD36+ and CD4+CD25–CD69+ cells in seronegative PR patients with active flares compared with healthy controls ($P < 0.0001$ for both cell subset comparisons). Gene enrichment analyses of RNA-sequencing data from sorted CD14+CD11b+CD36+ and CD4+CD25–CD69+ cells showed involvement of the inflammatory/stress response, phagocytosis, and regulation of apoptosis functional pathways. Up-regulated expression of *CXCL16* and *IL10RA* was observed in monocytes from PR patients. Up-regulation of *PFKFB3*, *DDIT4*, and *TGFB1*, and down-regulation of *PDIA6* were found in monocytes and lymphocytes from PR patients with active flares and PR patients in intercritical periods. Plasma levels of S100A8/A9 and interleukin-1 β were elevated in PR patients. Whole-exome sequencing revealed novel polygenic mutations in *HACL1*, *KDM5A*, *RASAL1*, *HAVCR2*, *PRDM9*, *MBOAT4*, and *JRKL*.

Conclusion. In seronegative PR patients, we identified a distinct CD14+CD11b+CD36+ cell subset that can induce an inflammatory response under stress and exert antiinflammatory effects after phagocytosis of apoptotic cells, and a CD4+CD25–CD69+ T cell subset with pro- and antiinflammatory properties. Individuals with genetic mutations involving epigenetic modification, potentiation and resolution of stress-induced inflammation/apoptosis, and a dysregulated endoplasmic reticulum stress response could be predisposed to seronegative PR.

INTRODUCTION

Palindromic rheumatism (PR) is a syndrome characterized by short relapsing and remitting episodes of painful joint swelling that

do not result in joint erosion (1,2). Although PR was first described by Hench and Rosenberg more than 75 years ago, the disease pathogenesis remains unclear (3). In the original cohort, PR was found to involve the rapid onset of periarticular inflammation,

Supported by China Medical University Hsinchu Hospital (grants CMUHCH-DMR-111-001 and CMUHCH-DMR-111-029), China Medical University Hospital (grants DMR-CELL-1901 and CMUH-DMR-104-112), and the Taiwan Ministry of Science and Technology (grant 110-2314-B-039-017).

¹Chin-An Yang, MD, PhD: College of Medicine, China Medical University, Division of Laboratory Medicine, China Medical University Hsinchu Hospital, and Departments of Medical Education and Pediatrics, China Medical University Hsinchu Hospital, Zhubei City, Taiwan, and Department of Biomedical Engineering and Environmental Sciences, National Tsing Hua University, Taiwan; ²Ju-Pi Li, PhD: Rheumatic Diseases Research Center, China Medical University Hospital, Department of Pathology, School of Medicine, Chung Shan Medical University and Chung Shan Medical University Hospital, Taiwan; ³Yi-Hua Lai, PhD, Joung-Liang Lan, MD: College of Medicine,

China Medical University, Rheumatic Diseases Research Center, China Medical University Hospital, and Rheumatology and Immunology Center, China Medical University Hospital, Taiwan; ⁴Ya-Ling Huang, MSc, Chien-Yu Lin, MSc: Division of Laboratory Medicine, China Medical University Hsinchu Hospital, Zhubei City, Taiwan.

Author disclosures are available at <https://onlinelibrary.wiley.com/action/downloadSupplement?doi=10.1002%2Fart.42297&file=art42297-sup-0001-Disclosureform.pdf>.

Address correspondence via email to Joung-Liang Lan, MD, at jounglan@me.com.

Submitted for publication December 10, 2021; accepted in revised form July 7, 2022.

potentially precipitated by stressors such as menses, mild infection, or overwork; the symptoms lasted for a few hours to several days and completely subsided between episodes (3). The clinical phenotype of intermittent self-limiting flares and the results of imaging studies showing extracapsular inflammation without synovitis during flares suggest that PR may be an autoinflammatory disorder (1,4,5).

The etiologies underlying PR seem to be heterogeneous. Besides its autoinflammatory features, a high percentage (40–60%) of PR patients in Western countries are positive for rheumatoid factor (RF) and anti-citrullinated protein antibodies (ACPAs)/anti-cyclic citrullinated peptide (anti-CCP) antibodies, and progression to rheumatoid arthritis (RA) has been observed in a substantial proportion (40–50%) of PR patients after many years of follow-up (4,6,7). PR has been suggested to be part of a continuum of RA with different autoimmune and autoinflammatory characteristics (2,7). However, the reported frequency of RF or anti-CCP antibody positivity in cohorts of Taiwanese patients with PR (<15%) is much lower than in cohorts of patients with PR in Western countries, and only ~12.9% of Taiwanese patients with PR developed RA during long-term follow-up (8–10). Therefore, we focused on a subgroup of patients with autoinflammatory PR who were seronegative for RF and anti-CCP antibodies in our study.

Distinct genetic causes associated with PR have been reported in Asian populations. A Korean study identified HLA-DRB1*0803 as a PR susceptibility gene; this gene has not been reported to be associated with any other rheumatic disease (11). Furthermore, variants in autoinflammation-related genes, such as tumor necrosis factor (TNF) receptor superfamily member 1A polymorphisms, were reported to be associated with PR in a Taiwanese cohort of mostly autoantibody-negative patients (10). Yet, the mechanisms underlying how certain polymorphisms and genes are linked to seronegative PR are unclear. Furthermore, the autoinflammatory immune phenotype, such as the key immune cell population and the plasma cytokine signature in seronegative PR, has not been characterized.

We aimed to investigate the autoinflammatory mechanisms underlying seronegative PR in a Taiwanese population. To this end, we used a multiomics approach involving mass cytometry immunophenotyping, RNA sequencing, and immune cell function assays to identify distinct CD14+CD11b+CD36+ and CD4+CD25–CD69+ immune cell subsets with potential roles in stress-induced autoinflammation, and performed whole-exome sequencing to reveal genetic variants that may predispose individuals to seronegative PR.

PATIENTS AND METHODS

Study design and participants. This was a prospective study recruiting Taiwanese patients at the Rheumatology and Immunology Center, China Medical University Hospital, Taichung,

Taiwan, from August 2019 to April 2022. Eligible patients were adults who fulfilled the PR diagnostic criteria proposed by Guerne and Weisman (7) and tested negative for RF and anti-CCP antibodies. None of the patients had a history of concurrent malignant disease or other rheumatic disease. Patients were excluded if their condition progressed to any rheumatic disease during the follow-up.

Patients were divided into 2 groups: PR patients having an acute flare of symptoms (active PR), and PR patients whose symptoms had subsided for at least 1 week (intercritical PR). Healthy controls without any rheumatic disease or related family histories were also recruited and matched with seronegative PR patients by similar ethnicity, sex, and age (see Supplementary Table 1 available on the *Arthritis & Rheumatology* website at <https://onlinelibrary.wiley.com/doi/art.42297>). For comparison, we recruited 5 PR patients in whom progression to RA had already been observed in prior analyses (2 patients were seronegative, 3 patients were seropositive for RF and anti-CCP antibodies). This study was approved by the Institutional Research Ethics Committee of the China Medical University Hospital, Taichung, Taiwan (approval nos. CMUH108-REC2-018, CMUH109-REC3-165, CMUH104-REC1-093). All participants provided written informed consent.

Sample collection. Blood samples were obtained from patients having an acute flare of symptoms (active PR) or from patients whose pain and swelling had subsided for at least 1 week (intercritical PR). Peripheral blood mononuclear cells (PBMCs) were isolated from venous whole blood collected from PR patients and healthy controls using Ficoll-Paque Plus (GE Healthcare) density-gradient centrifugation following the manufacturer's instructions and cryopreserved in a liquid nitrogen tank for future cytometry and RNA and DNA extraction. Plasma was collected from EDTA-anticoagulated whole blood and from heparinized whole blood after centrifugation for 10 minutes at 1,800g and stored at –80°C for subsequent enzyme-linked immunosorbent assay (ELISA) analyses according to the manufacturer's protocols.

Mass cytometry. Two million PBMCs were incubated with rhodium viability staining reagent (1:1,000 dilution; Standard BioTools) at 37°C for 15 minutes. After washing with Maxpar Cell Staining Buffer (Standard BioTools) at 300g for 5 minutes, the cells were resuspended in the same buffer and stained with a cytometry by time-of-flight mass spectrometry (CytoF) antibody cocktail (a combination of a Maxpar Human Monocyte/Macrophage Phenotyping Panel Kit and a Maxpar Human T-Cell Phenotyping Panel Kit, both from Standard BioTools) according to the manufacturer's instructions (see Supplementary Table 2 available on the *Arthritis & Rheumatology* website at <https://onlinelibrary.wiley.com/doi/art.42297>). A detailed protocol for staining and CytoF data acquisition is described in the

Supplementary Methods available at <https://onlinelibrary.wiley.com/doi/art.42297>.

The gating strategy used for performing mass cytometry analysis is shown in Supplementary Figure 1 available at <https://onlinelibrary.wiley.com/doi/art.42297>. High-dimensional data reduction and visualization analyses were performed using SPADE version 3.0 (12) (Supplementary Figure 2, <https://onlinelibrary.wiley.com/doi/art.42297>), and the t-distributed stochastic neighborhood embedding (tSNE) plugin in FlowJo version 10. High-dimensional data reduction and analyses are shown in the Supplementary Methods, <https://onlinelibrary.wiley.com/doi/art.42297>.

Flow cytometry analysis and fluorescence-activated cell sorting. PBMCs derived from PR patients and healthy controls were also prepared for flow cytometry analysis. Specifically, cells were stained with the following panels of fluorophore-conjugated antibodies: allophycocyanin-conjugated anti-CD14 (M5E2), fluorescein isothiocyanate (FITC)-conjugated anti-CD11b (ICRF44), and phycoerythrin (PE)-conjugated anti-CD36 (CB38); or FITC-conjugated anti-CD4 (RPA-T4), BB700-conjugated anti-CD69 (FN50), and PE-conjugated anti-CD25 (M-A251) (all from BD Biosciences). Stained cells were analyzed using a FACSLytic Flow Cytometry System (BD Biosciences). The percentages of CD14+CD11b+CD36+ cells and CD4+CD25-CD69+ cells were compared among healthy controls, active PR patients, and intercritical PR patients using Kruskal-Wallis tests. Distinct cell subsets were also sorted by fluorescence-activated cell sorting (FACS) with a FACSria III Cell Sorter (BD Biosciences). Sorted CD14+CD11b+CD36+ cells and remaining PBMCs, and sorted CD4+CD25-CD69+ cells and remaining cells in the lymphocyte gate from 12 PR patients were pooled into the following sample groups for subsequent RNA sequencing and differential gene expression analysis: 3 samples of the sorted CD14+CD11b+CD36+ cells, 3 samples of PBMCs depleted of CD14+CD11b+CD36+ cells, 3 samples of sorted CD4+CD25-CD69+ cells, and 3 samples of lymphocytes depleted of CD4+CD25-CD69+ cells.

RNA sequencing. Total RNA was extracted with TRIzol reagent (Cyrusbioscience) from sorted CD14+CD11b+CD36+ cells, PBMCs depleted of CD14+CD11b+CD36+ cells, sorted CD4+CD25-CD69+ cells, and lymphocytes depleted of CD4+CD25-CD69+ cells. Protocols for RNA sequencing and differential gene expression analysis are described in the Supplementary Methods available at <https://onlinelibrary.wiley.com/doi/art.42297>.

Real-time polymerase chain reaction. Two million PBMCs were incubated overnight in a 37°C incubator. The adherent cells were collected as the monocyte fraction, and the suspended cells were collected as the lymphocyte fraction.

Total RNA was extracted from both PBMC fractions with TRIzol reagent (Cyrusbioscience) and reverse transcribed using a cDNA PrimeScript Reverse Transcriptase Kit (Takara Bio). Protocols and primers for real-time polymerase chain reaction (PCR) are described in the Supplementary Methods available at <https://onlinelibrary.wiley.com/doi/art.42297>.

Determination of reactive oxygen species levels in PBMCs. Levels of reactive oxygen species (ROS) were determined using a DCFDA/H2DCFDA Cellular ROS Assay Kit (Abcam) per the manufacturer's instructions. Two million PBMCs were incubated overnight in a 37°C incubator. Then, the adherent monocytes were mixed with 20 μ M dichlorofluorescein diacetate (DCF-DA) for 30 minutes at 37°C in the dark. Next, the cells were stimulated for 1 hour with 1 μ M phorbol 12-myristate 13-acetate (PMA; MilliporeSigma). Finally, after washing, DCF-DA was detected with a FACSLytic Flow Cytometry System. Results are expressed as the percentage of PMA-stimulated DCF-DA-positive cells relative to baseline levels based on immunofluorescence staining.

Survival analysis of monocyte-derived macrophages. Monocytes were isolated by an overnight adhesion method and incubated in RPMI medium containing 10% autologous plasma (previously heat-inactivated at 56°C for 30 minutes) and 1% penicillin/streptomycin (Gibco, ThermoFisher) for 7 days. Afterward, the cells were stained with trypan blue (MilliporeSigma), and viable cells were counted using ImageJ version 1.53t.

Determination of plasma cytokine levels using ELISA. Plasma was collected from anticoagulated whole blood after centrifugation for 10 minutes at 1,800g. Plasma cytokine levels were determined with ELISA kits for interleukin-1 β (IL-1 β), S100A8/A9, TNF, and transforming growth factor β 1 (TGF β 1) (all from R&D Systems) and for S100A12 (Abcam) according to the manufacturers' instructions.

Whole-exome sequencing. Genomic DNA was extracted from PBMCs from 3 unrelated patients with seronegative PR using a DNeasy Blood & Tissue Kit (Qiagen) per the manufacturer's guidelines. Whole-exome sequencing libraries were prepared using KAPA HyperPlus Kits with KAPA HyperCapture Reagent Kits (both from Roche) according to the manufacturer's protocols. Further steps for whole-exome sequencing library preparation, sequencing method, data annotation, and criteria for pathogenic variant selection are presented in the Supplementary Methods available at <https://onlinelibrary.wiley.com/doi/art.42297>.

Statistical analysis. Data from analyses of immune cell subsets or cytokine levels in which 3 groups were compared were

analyzed using Kruskal-Wallis nonparametric tests in GraphPad Prism version 8. Mann-Whitney U tests were used for comparisons between 2 groups. Statistical significance was defined as a *P* value less than 0.05. The Benjamini-Hochberg method was used to correct *P* values for multiple comparisons.

RESULTS

Patients and clinical characteristics. A total of 57 PR patients who fulfilled the Guerne and Weisman criteria (7) were screened. Six patients were seropositive for RF or anti-CCP and were excluded. Fifty-one PR patients who were seronegative for RF and anti-CCP were recruited and followed up from August 2019 to April 2022; however, 3 PR patients were further excluded due to their condition progressing to RA during follow-up (fulfilled the 2010 American College of Rheumatology [ACR]/EULAR classification criteria for RA) (13). The resulting cohort included 48 seronegative PR patients without development of other rheumatic diseases and 48 healthy controls (see Supplementary Figure 3 available on the *Arthritis & Rheumatology* website at <https://onlinelibrary.wiley.com/doi/art.42297>).

The demographic features and clinical characteristics of the seronegative PR patients are presented in Supplementary Table 1, <https://onlinelibrary.wiley.com/doi/art.42297>. There was no statistically significant difference in sex or age between the seronegative PR patients and healthy controls. The median disease duration was 9 years (range 3–40 years), and the median period of time between flares, counted as the number of days between the last 2 flares at the time of blood sampling, was 20 days (range 5–150 days). No erosion was detected in joint imaging studies using ultrasonography or radiography in 28 PR patients who had >1 flare per month during follow-up. Clinical features were similar between male and female patients.

Immunophenotyping of PBMCs derived from seronegative PR patients. We assessed whether there were differences in the immunophenotypes of PBMCs derived from PR patients versus healthy controls using a mass cytometry panel of 22 metal-tagged antibodies (Supplementary Table 2, available at <https://onlinelibrary.wiley.com/doi/art.42297>). As shown in tSNE graphs of concatenated data (Figures 1A–C), increased percentages of CD14+CD11b+CD36+ cells and CD4+CD25–CD69+ cells were detected in the PR groups compared with healthy controls (CytoF results gated on these 2 subsets are shown in Figures 1D–I and Supplementary Figures 4 and 5, <https://onlinelibrary.wiley.com/doi/art.42297>).

For comparison, we performed additional flow cytometry immunophenotyping of these 2 cell subsets on more PR patients and on 5 patients whose condition had previously progressed from PR to RA (fulfilled the 2010 ACR/EULAR classification criteria for RA) (Figures 1J and K). Compared with healthy controls,

markedly higher percentages of CD14+CD11b+CD36+ cells were detected in the active and intercritical PR groups (median percentage of CD14+CD11b+CD36+ cells, 3.24% in active PR versus 3.59% in intercritical PR versus 1.44% in healthy controls; *P* for trend = 0.0018 by Kruskal-Wallis test). However, the percentage of CD14+CD11b+CD36+ cells was lower in patients whose condition had progressed to RA compared with seronegative PR patients with active flares (median percentage of CD14+CD11b+CD36+ cells, 3.24% in active PR versus 0.41% in RA progressors; *P* = 0.0194 by Mann-Whitney U test) (Figure 1J). Furthermore, a markedly increased percentage of CD4+CD25–CD69+ cells was also observed in PR patients (median percentage of CD4+CD25–CD69+ cells, 4.67% in active PR versus 2.02% in intercritical PR versus 1.64% in healthy controls; *P* for trend = 0.0009 by Kruskal-Wallis test) (Figure 1K). A trend of lower percentages of CD4+CD25–CD69+ cells was detected in PBMCs derived from RA progressors compared with seronegative PR patients with active flares (median percentage of CD4+CD25–CD69+ cells, 4.67% in active PR versus 2.22% in RA progressors, *P* = 0.0532 by Mann-Whitney U test) (Figure 1K).

Together, these data reveal that there were immunophenotypic differences between seronegative PR patients and healthy controls, specifically with regard to increased proportions of CD14+CD11b+CD36+ and CD4+CD25–CD69+ cells in the seronegative PR patients. In contrast, an increase in these 2 cell subsets was not observed in patients whose condition had progressed to RA.

Enriched pathways of genes up-regulated in sorted CD14+CD11b+CD36+ cells and sorted CD4+CD25–CD69+ cells. To evaluate the distinct gene expression in the identified CD14+CD11b+CD36+ cells versus the remaining PBMCs and the distinct gene expression in CD4+CD25–CD69+ cells versus the remaining lymphocytes, we isolated these 4 subsets from seronegative PR patients to perform an RNA-sequencing analysis. Compared with the respective depleted cell populations, 12 genes (*THBS1*, *CD300E*, *HMOX1*, *CXCL16*, *PPIF*, *NAMPT*, *NLRP3*, *TRIB1*, *INSIG1*, *JARID2*, *C15orf39*, *IRF1*) were significantly up-regulated in the CD14+CD11b+CD36+ cell subset, and 2 genes (*DDIT4*, *PFKFB3*) were enriched in the CD4+CD25–CD69+ lymphocytes (Supplementary Table 3 and Supplementary Figure 6, available at <https://onlinelibrary.wiley.com/doi/art.42297>).

In addition, we analyzed the expression of these genes in sorted CD14+CD11b+CD36+ and CD4+CD25–CD69+ cells collected from 4 seropositive PR patients (Supplementary Table 4, <https://onlinelibrary.wiley.com/doi/art.42297>). A network analysis of these molecules using gene ontology (GO) annotation identified enriched pathways, including the defense response (GO:0006952), response to stress (GO:0006950), inflammatory response (GO:0006954), positive regulation of cell death (GO:0010942), phagocytosis and engulfment (GO:0006911),

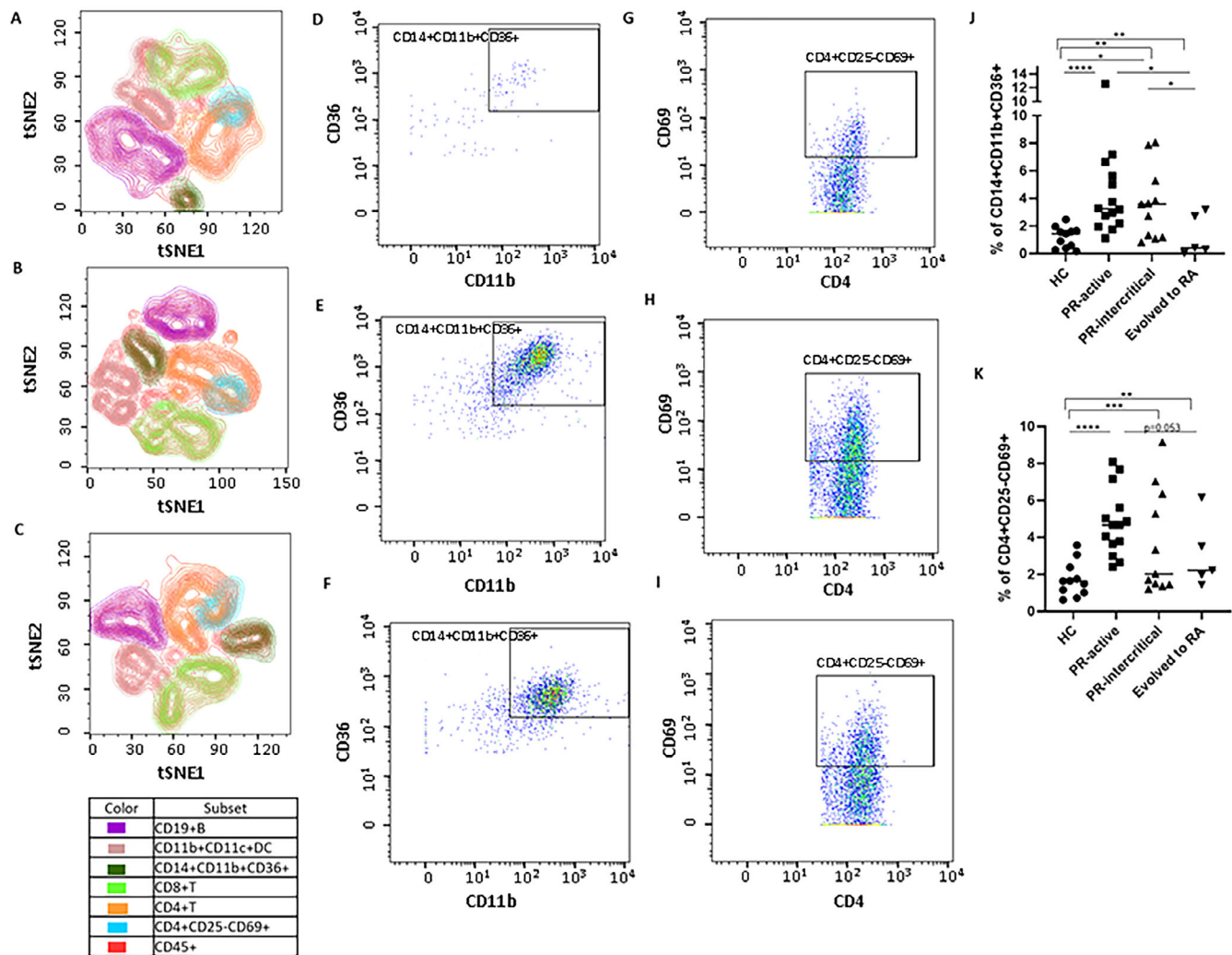


Figure 1. Immunophenotyping of peripheral blood mononuclear cells (PBMCs) from palindromic rheumatism (PR) patients seronegative for rheumatoid factor and anti-cyclic citrullinated peptide antibodies and healthy controls (HCs). **A–C**, Gating strategy using t-distributed stochastic neighborhood embedding (tSNE) with FlowJo version 10 for mass cytometry analysis of cell lineages (color coded) among PBMCs from HCs ($n = 5$) (**A**), PR patients with active flares ($n = 8$) (**B**), and PR patients in an intercritical period without active symptoms ($n = 5$) (**C**). **D–I**, Cytometry by time-of-flight mass spectrometry analysis of PBMCs from HCs (**D** and **G**), active PR patients (**E** and **H**), and intercritical PR patients (**F** and **I**) showing gating of CD14+CD11b+CD36+ cells (among CD14+CD36+ cells) (**D–F**) and CD4+CD25-CD69+ cells (among CD4+CD25- cells) (**G–I**). **J** and **K**, Percentages of CD14+CD11b+CD36+ cells (**J**) and CD4+CD25-CD69+ cells (**K**) detected by flow cytometry in PBMCs from 11 HCs, 14 active PR patients, 11 intercritical PR patients, and 5 patients whose condition progressed to rheumatoid arthritis (RA). In **J** and **K**, symbols represent individual samples; horizontal lines represent median percentages. * = $P < 0.05$ and **** = $P < 0.0001$ comparing 2 groups by Mann-Whitney U test; ** = $P < 0.01$ and *** = $P < 0.001$ comparing 3 or 4 groups by Kruskal-Wallis test.

and apoptosis process (GO:0006915) (Supplementary Figure 7, <https://onlinelibrary.wiley.com/doi/art.42297>).

Differential gene expression in monocytes and lymphocytes from seronegative PR patients. To investigate if the genes identified to be most enriched in CD14+CD11b+CD36+ cells or CD4+CD25-CD69+ cells are also differentially expressed in monocytes or lymphocytes from seronegative PR patients compared to healthy controls, we further performed real-time PCR analyses. Of the top 5 up-regulated genes identified in CD14+CD11b+CD36+ cells, in

addition to *NLRP3*, the key molecule for inflammasome activation, only *CXCL16* showed significant differential expression among monocytes derived from healthy controls, active PR patients, and intercritical PR patients (median relative *CXCL16* expression, 29.93-fold in active PR versus 46.48-fold in intercritical PR versus 4.95-fold in healthy controls; P for trend = 0.0045 by Kruskal-Wallis test, adjusted $P = 0.0083$) (Figures 2A–F). Compared with healthy controls, the expression levels of the endoplasmic reticulum (ER) stress response molecule *DDIT4* and a regulator of glucose metabolism, *PFKFB3*, were markedly increased in monocytes and lymphocytes from seronegative PR

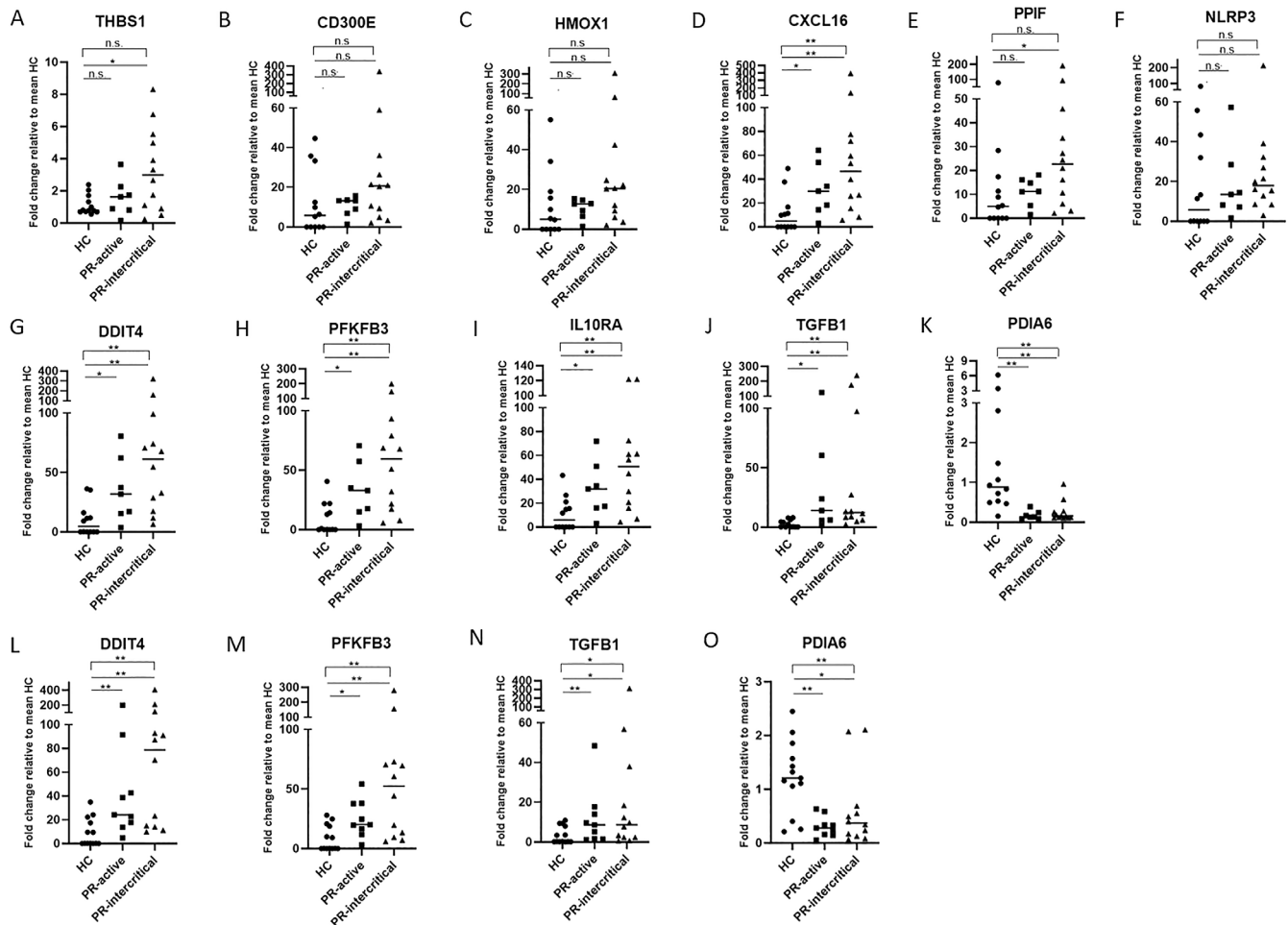


Figure 2. Differential gene expression determined by real-time polymerase chain reaction in monocytes (A–K) and lymphocytes (L–O) from PR patients seronegative for rheumatoid factor and anti-cyclic citrullinated peptide antibodies, according to those with active disease flares and those without symptoms in an intercritical period, compared with HCs. In A–K, monocytes were derived from 12 HCs, 7 active PR patients, and 12 intercritical PR patients. In L–O, lymphocytes were derived from 13 HCs, 9 active PR patients, and 12 intercritical PR patients. Symbols represent individual samples; horizontal lines show the median fold change relative to mean values in HCs. * = adjusted $P < 0.05$ and ** = adjusted $P < 0.01$ by Mann-Whitney U test in 2-group comparisons or by Kruskal-Wallis test in 3-group comparisons. n.s. = not statistically significant (see Figure 1 for other definitions).

patients (median relative *DDIT4* expression in monocytes, 31.78-fold in active PR versus 61.11-fold in intercritical PR versus 4.63-fold in healthy controls; P for trend = 0.0018 by Kruskal-Wallis test, adjusted $P = 0.0061$) (Figures 2G, H, L, and M).

Since genes enriched in the distinct CD14+CD11b+CD36+ and CD4+CD25–CD69+ cell subsets are involved in stress and inflammatory responses, we also analyzed the expression of the antiinflammatory molecules *IL10RA* and *TGFB1*, and the ER stress regulator *PDIA6* in monocytes and lymphocytes from seronegative PR patients (Figures 2I, J, K, N, and O). Elevated *TGFB1* and markedly decreased *PDIA6* expression levels were observed in monocytes and lymphocytes from seronegative PR patients compared with healthy controls (median relative *TGFB1* expression in monocytes, 14.07-fold in active PR versus 12.31-fold in intercritical PR versus 1.46-fold in healthy controls, P for trend = 0.0012 by Kruskal-Wallis test, adjusted $P = 0.0061$; median relative *PDIA6* expression in monocytes, 0.13-fold versus

0.16-fold versus 0.88-fold, respectively, P for trend = 0.0004 by Kruskal-Wallis test, adjusted $P = 0.0044$) (Figures 2J and K). The difference in *PDIA6* gene expression suggests dysregulated ER stress response as part of the disease mechanisms.

Elevated plasma proinflammatory cytokine levels in PR patients. To further characterize stress-induced immune dysregulation and cell death in PR, we measured levels of proinflammatory cytokines (S100A8/A9, S100A12, IL-1 β , TNF) in plasma obtained from seronegative patients with active PR and seronegative patients with intercritical PR (Figures 3A–D). Compared with healthy controls, PR patients had elevated levels of S100A8/A9 (median concentration, 1,724.18 ng/ml in active PR versus 1,647.84 ng/ml in intercritical PR versus 797.11 ng/ml in healthy controls; P for trend = 0.006 by Kruskal-Wallis test, adjusted $P = 0.015$) (Figure 3A) and IL-1 β (median concentration, 0.50 pg/ml in active PR versus 0.48 pg/ml in intercritical PR versus

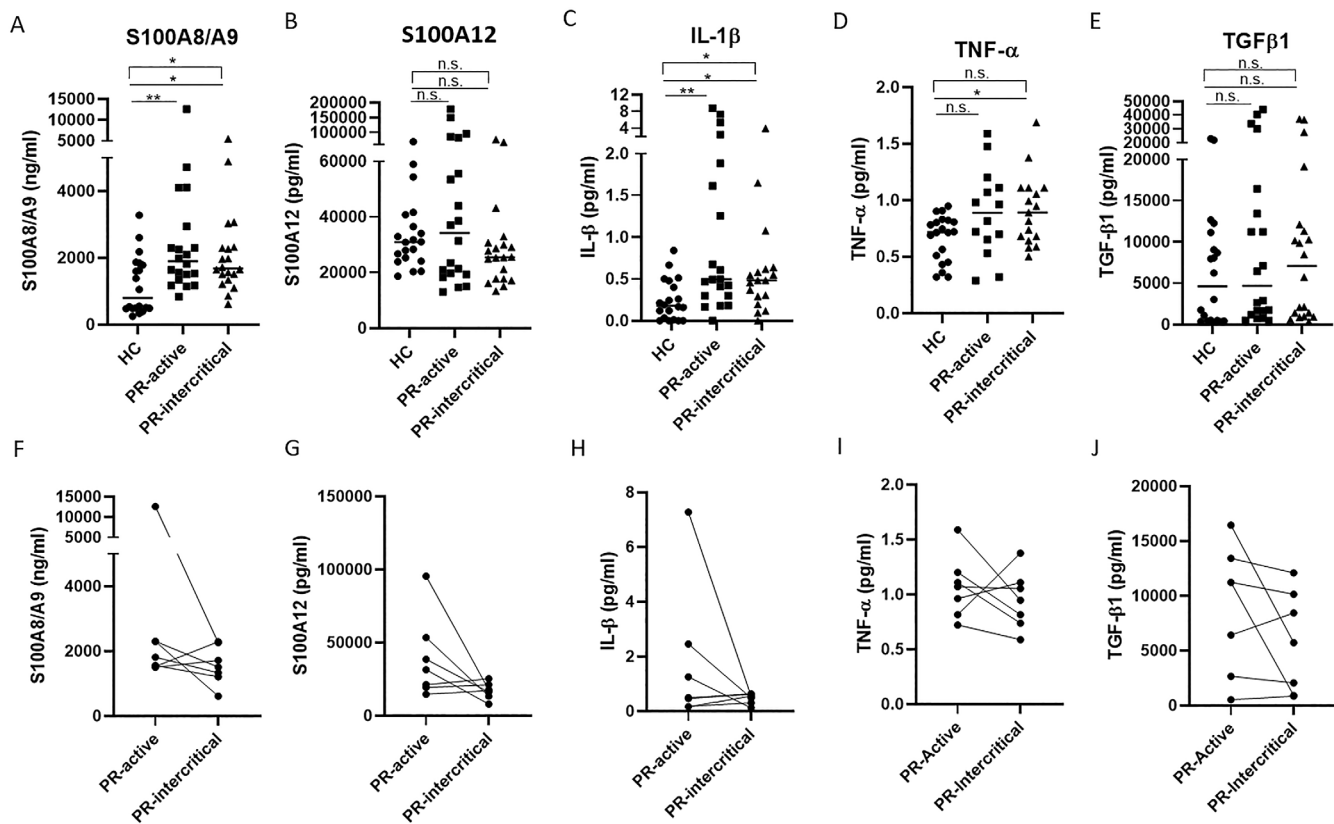


Figure 3. Cytokine levels in the plasma from PR patients seronegative for rheumatoid factor and anti-cyclic citrullinated peptide antibodies, according to those with active flares and those without symptoms during an intercritical period, compared with HCs (A–E), and in paired plasma samples (symbols with connecting lines) collected from the same seronegative PR patients ($n = 7$) during an active flare and during an intercritical period (F–J). In A–E, symbols represent individual samples; horizontal lines represent median values. * = adjusted $P < 0.05$ and ** = adjusted $P < 0.01$ by Mann-Whitney U test in 2-group comparisons and by Kruskal-Wallis test in 3-group comparisons. n.s. = not statistically significant; IL-1 β = interleukin-1 β ; TNF- α = tumor necrosis factor; TGF β 1 = transforming growth factor β 1 (see Figure 1 for other definitions).

0.18 pg/ml in healthy controls; P for trend = 0.0042 by Kruskal-Wallis test, adjusted $P = 0.0015$ (Figure 3C). Similar levels of the regulatory cytokine TGF β 1 were detected in plasma of PR patients and healthy controls (Figure 3E). We also evaluated the change in cytokine levels in plasma obtained from 7 PR patients during an active flare compared to plasma obtained during an intercritical period, and found that the 3 individuals with the highest cytokine levels during an active flare had decreased cytokine levels during an intercritical period (Figures 3F–J).

ROS production and number of apoptotic macrophages in PR patients. Since S100 proteins are known to be involved in macrophage inflammatory responses (14), we measured ROS production from monocytes separated from PBMCs from healthy controls and PR patients. Although high levels of stimulated ROS production were observed in several PR patients, there was no significant difference in induced percentages of DCF-DA-positive cells among the 3 groups (healthy controls, active PR, intercritical PR) (Supplementary Figure 8A, <https://onlinelibrary.wiley.com/doi/art.42297>).

To evaluate the frequency of inflammation-induced cell death in PR patients, we derived macrophages from monocytes of healthy controls and monocytes of PR patients by adding autologous plasma to the culture medium. In comparing the percentage of dead cells in monocyte-derived macrophages, fewer dead cells were seen among monocyte-derived macrophages from patients with intercritical PR compared with monocyte-derived macrophages from patients with active PR (median percentage of cell death, 51.10% in intercritical PR patients versus 72.17% in active PR patients; $P = 0.030$). Moreover, a lower percentage of dead cells was seen among monocyte-derived macrophages from patients with intercritical PR compared with monocyte-derived macrophages from healthy controls (median percentage of cell death, 51.10% in intercritical PR patients versus 88.57% in healthy controls; $P = 0.0087$) (Supplementary Figure 8B, <https://onlinelibrary.wiley.com/doi/art.42297>).

To investigate whether the use of hydroxychloroquine (HCQ) might influence the level of inflammation and contribute to the difference observed between healthy controls and PR patients, we analyzed the effect of HCQ on lipopolysaccharide-induced inflammatory response in human monocytic cell line THP-1.

RA (1 female/2 male patients, mean $\pm$ SD age 30 ± 9.5 years, mean $\pm$ SD disease duration 8 ± 4.0 years, 1 patient taking nonsteroidal antiinflammatory drugs, 2 patients taking HCQ). Genetic mutations in *HACL1*, *KDM5A*, *RASAL1*, *PRDM9*, *HAVCR2*, *MBOAT4*, and *JRKL* were detected, and the predicted impacts of these genetic variants on functional protein domains are shown in Figure 4.

Further network and gene enrichment analyses of the identified pathogenic genes, together with molecules up-regulated in CD14+CD11b+CD36+ and CD4+CD25-CD69+ cells, and plasma

of seronegative PR. To investigate the potential genetic causes of seronegative PR, we performed whole-exome sequencing analyses of samples from 3 unrelated patients with no family history of

Figure 4. Genetic mutations detected using whole-exome sequencing in 3 palindromic rheumatism patients (proband 1, proband 2, proband 3) who were seronegative for rheumatoid factor and anti-cyclic citrullinated peptide antibodies. The Simple Modular Architecture Research Tool (SMART, smart.embl.de) was used to analyze the protein domains potentially affected by the identified pathogenic genetic variants (**blue arrows**). Sanger sequencing results validating the identified variants show the sites of the mutations (**red arrows**). In proband 1, heterozygous mutations were detected in *HACL1* (c.1506_1510dup, predicted to affect the functional Pfam domain), *KDM5A* (c.485A>G, predicted to affect the functional JmjC domain), and *RASAL1* (c.2066G>T, predicted to affect the functional BTK domain). In proband 2, a homozygous mutation was detected in *PRDM9* (c.2125_2126insGTCAACCAGAGGACACACACAGGGGAGAAGCCCTATGTCTGCACGGGAGTGTGGGCGGGGCTTTAGCTGGCAGT-CAGTCCTCCTCA, predicted to affect the functional ZnF_C2H2 domain), and a heterozygous mutation was detected in *HAVCR2* (c.344dup, predicted to affect the functional IG domain). In proband 3, heterozygous mutations were detected in *MBOAT4* (c.1038del, predicted to affect the functional Pfam domain) and *JRKL* (c.621_624del, predicted to affect the functional DDE_1 domain).

from seronegative PR patients were done using GO analysis with STRING version 11.5. The analyses revealed that the detected genetic variants are involved in the following GO annotations: response to stress (GO:0006950), positive regulation of NF-κB transcription factor activity (GO:0051092), cellular response to oxidative stress (GO:0034599), regulation of apoptotic signaling pathway (GO:2001233), inflammatory response (GO:0006954), cell death (GO:0008219), stress-activated MAPK cascade (GO:0051403), activation of MAPK activity (GO:0000187), Toll-like receptor (TLR) signaling pathway (GO:0002224), and positive regulation of ROS metabolic process (GO:2000379) (Figure 5).

DISCUSSION

In this study, we performed immunophenotyping, gene expression, and genetic mutation analyses of blood samples obtained from PR patients seronegative for RF and anti-CCP

antibodies. Seronegative PR patients are the main PR patient subpopulation at our clinic (RF or anti-CCP seropositive rate, 10.53% [6 of 57 patients] at screening) and in 2 other reported cohorts of Taiwanese PR patients (RF or anti-CCP seropositive rate, 14.29% in PR and 5.36% in persistent PR without progression to RA, respectively) (8,10). Similar to findings from previous studies of Taiwanese PR patients (8,10), RF and anti-CCP seronegative PR patients in our cohort were less likely to experience progression of PR to RA (5.88% [3 of 51 patients]) compared with seropositive PR patients (50% [3 of 6 patients]) over follow-up. Therefore, we focused specifically on characterizing the autoinflammatory mechanisms underlying this distinct seronegative PR group (persistently seronegative for RF and anti-CCP antibodies without progression of PR to RA).

We found markedly elevated percentages of CD14+ CD11b+CD36+ cells and CD4+CD25–CD69+ cells among PBMCs derived from seronegative PR patients. RNA sequencing

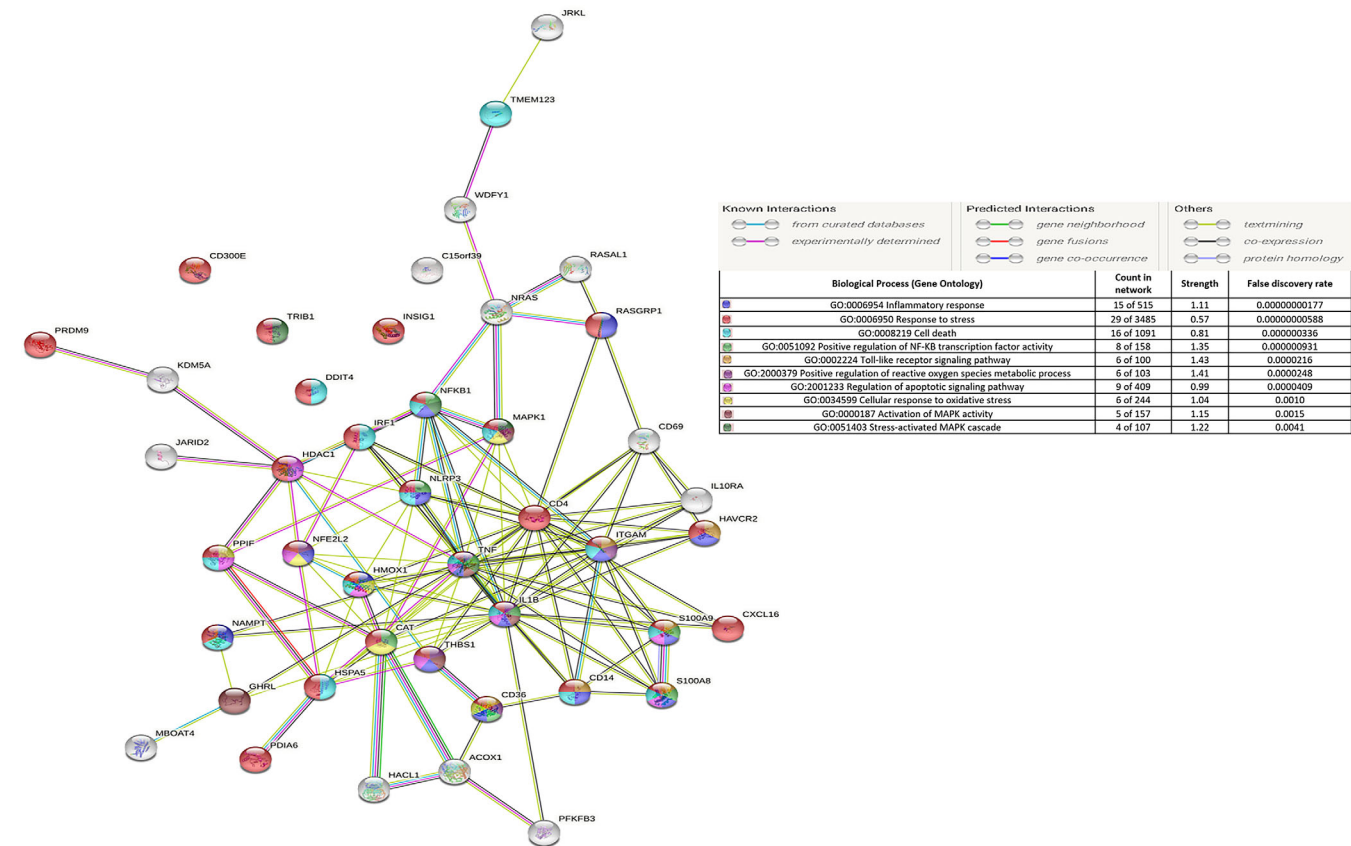


Figure 5. Network and pathway enrichment analyses of the identified pathogenic genes in 3 palindromic rheumatism patients seronegative for rheumatoid factor and anti-cyclic citrullinated peptide antibodies using gene ontology (GO) analysis with STRING version 11.5. The key defines the molecules enriched in specific pathways (color nodes) and the types of interactions (connecting lines). Among the protein network analysis, *HACL1* was found to have close interaction with *CAT* (combined interaction score 0.955), which is involved in pathways of response to stress, positive regulation of NF-κB transcription factor activity, and cellular response to oxidative stress. Furthermore, *KDM5A* was found to have potential interactions with *PRDM9* (combined interaction score 0.402) and *HDAC1* (combined interaction score 0.943), a key molecule involving pathways of stress response and regulation of apoptosis. *HDAC1* also had potential interaction with *NFE2L2* (NF-E2-related factor 2 [Nrf2]), a transcription factor binding to the antioxidant response element. *HAVCR2* was found to be involved in pathways of inflammatory response, stress response, and Toll-like receptor signaling. *MBOAT4* and *RASAL1* were among molecules enriched in the pathways of activation of MAPK activity and cell death, respectively, and had potential interactions with molecules linked to immune activation.

of the sorted CD14+CD11b+CD36+ cell subset suggested these cells have a phagocytic role in TLR/stress-induced immune activation and apoptotic cell clearance. RNA sequencing of the sorted CD4+CD25–CD69+ lymphocytes suggested they have a role in regulating stress response. Increased levels of stress-related cytokines were also detected in plasma from seronegative PR patients. Whole-exome sequencing of seronegative PR patients revealed novel polygenic mutations in *HACL1*, *KDM5A*, *RASAL1*, *HAVCR2*, *PRDM9*, *MBOAT4*, and *JRKL*.

The identified CD14+CD11b+CD36+ cell subset enriched in PBMCs of seronegative PR is a distinct myeloid cell type which has not been previously described in blood. The CD14+CD11b+CD36+ cells found in seronegative PR patients could have both proinflammatory and antiinflammatory behaviors. While CD14 is a well-known monocyte marker, CD11b has been shown to act in concert with CD14 to activate TLR-4 signaling (15). DAMPs, such as S100 proteins, can stimulate phagocytes to secrete proinflammatory cytokines via TLR-4 or receptor for advanced glycation end products signaling (16,17). Furthermore, CD36 expressed on macrophages has been reported to recognize apoptotic cells via interactions with integrins and thrombospondin 1 (TSP-1) (18). The CD36- $\alpha_v\beta_3$ -TSP-1 complex can stimulate TGF β 1 release to suppress inflammatory monocyte/macrophage responses upon encountering ACAMPs (19). It has been reported that although clearance of apoptotic cells usually exerts antiinflammatory effects on monocytes/macrophages via IL-10 or TGF β 1, certain conditions within the microenvironment of the phagocytic synapse allow apoptotic cells to stimulate the release of proinflammatory cytokines from monocytes/macrophages (19).

Indeed, enriched expression of the CD36 ligand, TSP-1 (*THBS1*), along with other genes related to defense response and stress response (*CD300E*, *HMOX1*, *CXCL16*, *IRF1*, *TRIB1*, *INSIG1*), genes involving inflammatory response (*NAMPT*, *NLRP3*), and *PPIF*, a gene related to oxidative stress-induced cell death (20), were detected in the sorted CD14+CD11b+CD36+ cells. Of note, in addition to activating macrophages and providing survival signals, myeloid cell surface CD300e can also decrease HLA class II expression and down-regulate STAT1-mediated antigen-specific T cell expansion (21). Similarly, *HMOX1*, which encodes heme oxygenase 1, has been reported to inhibit oxidative damage and apoptosis, and to mediate IL-10 antiinflammatory signaling upon stress, promoting the resolution of inflammation (22). Furthermore, it has been shown that *CXCL16* expression might restrain stimulated microglia/macrophage polarization toward an inflammatory phenotype (23,24). Therefore, the distinct CD14+CD11b+CD36+ cell subset might evoke proinflammatory responses during acute stress yet have antiinflammatory effects after the phagocytosis of apoptotic cells, ultimately shaping the phagocytic synapse toward a microenvironment that favors the resolution of inflammation.

In our immunophenotyping experiments, we also detected an increased frequency of CD4+CD25–CD69+ T cells in seronegative PR patients. CD4+CD25–CD69+ T cells were previously discovered as a new subset of regulatory T cells that suppress T cell proliferation via membrane-bound TGF β 1 (25). As shown in Supplementary Figures 10A and B (available at <https://onlinelibrary.wiley.com/doi/art.42297>), we detected cell surface TGF β 1 expression on CD4+CD25–CD69+ cell subsets, but not on CD4+CD25–CD69– cell subsets. Moreover, there is evidence that dendritic cells that ingest apoptotic cells have a reduced capacity to stimulate T cell activation (19), and human CD36^{high} monocytes have been previously shown to induce regulatory FOXP3+CD25+ T cells from CD4+ and CD8+ cell subsets (26). Therefore, it is possible that the identified CD14+CD11b+CD36+ cell subset not only develops reduced inflammatory and possibly immunosuppressive potential through phagocytosis of apoptotic cells, but also induces the regulatory CD4+CD25–CD69+ T cells.

Consistent with this hypothesis, the expression of *DDIT4* and *PFKFB3* were found not only to be enriched in sorted CD4+CD25–CD69+ lymphocytes but also up-regulated in lymphocytes and monocytes from PR patients compared with healthy controls. *DDIT4* has been reported to be induced by ER stress, and its up-regulation in stimulated T cells promotes T cell proliferation and survival (27,28). *DDIT4* is a negative regulator of metabolic stress, which can inhibit mechanistic target of rapamycin complex 1 (mTORC1)-mediated apoptosis induced by ER stress (29,30). Since mTOR is the key molecule integrating activation signals in mature, naive T cells, *DDIT4* might modulate the type of T cell activation and differentiation via inhibition of mTORC1. The glycolytic enzyme *PFKFB3* also has pro- and antiinflammatory functions. While *PFKFB3* has been reported to potentiate LPS-triggered NF- κ B inflammation, reduced *PFKFB3* expression observed in T cells of RA patients lead to redox imbalance and consumption of intracellular ROS, which could augment the differentiation of T cells into the proinflammatory Th17 cells (31–33). These results support the evidence that CD4+CD25–CD69+ cells are activated T cells with immune-regulatory functions.

Of note, the markedly increased percentages of CD14+CD11b+CD36+ cells and CD4+CD25–CD69+ cells detected in PBMCs of seronegative active PR patients were not observed in 5 PR patients who had recently experienced progression of PR to RA (fulfilled the 2010 ACR/EULAR RA classification criteria; 2 seronegative, 3 seropositive for RF and anti-CCP), suggesting the etiology of seronegative PR might be different from the mechanisms leading to RA development. Due to the small sample size of the seropositive PR population in our cohort, we could only analyze the gene expression patterns in a limited number (not enough for differential gene expression analysis) of sorted CD14+CD11b+CD36+ and CD4+CD25–CD69+ cells for comparison. The expression level of *THBS1* seemed to be higher in

CD14+CD11b+CD36+ cells derived from seropositive PR patients compared with the same cell subset from seronegative PR patients (Supplementary Table 4, available at <https://onlinelibrary.wiley.com/doi/art.42297>). Although TSP-1 has been reported to be up-regulated in RA and play a proinflammatory role (34), further experiments need to be done to investigate the mechanistic difference between seronegative PR and seropositive PR in the Taiwanese population.

In monocytes and lymphocytes isolated from seronegative PR patients, we investigated the expression of the genes identified to be most enriched in CD14+CD11b+CD36+ cells or CD4+CD25-CD69+ cells. As described in the above paragraph, stress-induced molecules with pro- and antiinflammatory roles (*CXCL16*, *DDIT4*, and *PFKFB3*) were up-regulated both in seronegative patients with active PR and in seronegative patients with intercritical PR compared with healthy controls. In addition, the antiinflammatory molecules *IL10RA* and *TGFB1* were also up-regulated in monocytes and lymphocytes from seronegative PR patients.

In contrast, the expression of the ER stress regulator *PDIA6* decreased significantly in monocytes and lymphocytes of seronegative PR patients compared with healthy controls. Low-grade inflammation, which is a potential trigger for a PR flare, can disrupt protein folding in the ER, the key organelle for this process and protein transport. ER stress further activates the adaptive ER unfolded protein response to restore ER homeostasis (35). Interestingly, unfolded protein response signaling intersects with inflammatory pathways, including the NF- κ B/IKK and JNK/AP-1 pathways (36). ER stress cannot be reversed if the unfolded protein response fails, and the accumulation of unfolded proteins can lead to cell death and inflammation; the inability to restore ER homeostasis has been shown to play important roles in multiple inflammatory and metabolic diseases (35).

Our findings of enriched expression of genes related to ER stress and decreased expression of *PDIA6*, which regulates the activation of the key unfolded protein response-signaling molecule inositol-requiring enzyme 1 α (IRE-1 α) (37), suggest dysregulated unfolded protein response upon ER stress in seronegative PR patients. Further investigations on the expression levels of other downstream molecules of unfolded protein response signaling, including IRE-1 α , PERK, and activating transcription factor 6, in seronegative PR patients are needed to characterize the dysregulated ER stress response.

Elevated levels of S100A8/A9 and IL-1 β detected in plasma of seronegative PR patients provide evidence of stress-induced inflammation as a mechanism underlying PR flares. Higher levels of S100A8/A9 have been reported to reflect neutrophil activation and have been correlated with disease activity in other inflammatory rheumatic diseases, such as RA and axial spondyloarthritis (38). In seronegative PR, however, levels of S100A8/A9 and IL-1 β were not associated with active or intercritical PR. Active flares and intercritical periods of seronegative PR were better differentiated by the level of serum C-reactive protein in our

cohort. It should be noted that most of our experiments were done in unpaired samples; therefore, we could not determine the exact change of cytokine levels between active and intercritical PR periods in the same individual. Furthermore, the inflammatory response measured in PR patients might be influenced by the use of HCQ when compared with healthy controls.

One limitation of our study was that not all patients enrolled in this study were treatment-naïve. Most of the patients had been treated with HCQ, a suggested first-line therapy for PR (39). Although the mechanisms of action of HCQ in seronegative PR are unclear, and randomized controlled trials evaluating the therapeutic efficacy of the drug are still lacking, HCQ has been shown to inhibit IL-1 β production in amyloid-stimulated neutrophils, and chloroquine has been demonstrated to inhibit IL-1 β , TNF, and IL-6 secretion in LPS-stimulated monocytes (40,41). In our experiments, pretreatment with HCQ decreased LPS-induced IL-1 β and ROS production in THP-1 cells. These results suggest that IL-1 β and ROS levels could be higher in PR patients not receiving HCQ treatment and partly explain why we did not detect a difference of stimulated monocyte ROS production in PR patients compared with healthy controls. In addition, the effect of HCQ on excessive ER stress-induced cell death is unclear. In our study, the decreased death rate of macrophages derived from monocytes of intercritical PR patients compared with active PR patients and decreased C-reactive protein levels in intercritical PR patients suggest signs of inflammation resolution.

Whole-exome sequencing of samples from 3 subjects with seronegative PR revealed multiple potentially pathogenic genetic variants that could be correlated with the disease. The 3 patients had variants in *HACL1*, *KDM5A*, and *RASAL1*; *PRDM9* and *HAVCR2*; or *MBOAT4* and *JRKL*. In our network analysis, *HACL1* was found to interact with the antioxidant enzyme catalase CAT, which had potential interaction with Hsp70 member 5 (HSPA5), a master protein in the ER unfolded protein response process. *KDM5A*, an epigenetic regulator, was found to have potential interactions with *PRDM9* and *HDAC1*. Reduced *Hdac1* protein in rats has been shown to result in prolonged nuclear IL-1 β -induced NF- κ B p65 protein phosphorylation (42). *MBOAT4* converts ghrelin to the active form acyl ghrelin. The ghrelin/growth hormone secretagogue receptor 1a system has been reported to counteract the stimulatory effects of IL-1 β on IL-6 and IL-8 production (43).

Genetic variants affecting immune effector functions, such as those in *RASAL1* and *HAVCR2*, might further potentiate stress-induced inflammation and cell death. In addition, *HAVCR2* encodes T cell immunoglobulin and mucin domain-containing protein 3 (TIM-3), which mediates the phagocytic clearance of apoptotic cells (44). A loss-of-function mutation in the TIM-3 immunoglobulin V domain has been reported to cause macrophage hyperactivation and the secretion of inflammatory cytokines such as TNF and IL-1 β (45,46). The heterozygous *HAVCR2* c.344dup (p.Gly116ArgfsTer5) mutation found in 1

patient was predicted to affect the immunoglobulin domain, possibly causing impaired apoptotic cell clearance and phagocyte hyperactivity. Furthermore, *HAVCR2* and *JRKL* mutations have been reported to be associated with inflammatory bowel disease (47,48), in which dysregulated ER homeostasis plays a role (49). Therefore, we hypothesized that under minor stress, PR patients are more likely to accumulate excessive unfolded proteins via multiple mechanisms, triggering apoptosis and inflammation. Since long-lasting epigenetic remodeling is associated with changes in activating histone marks, including H3K27 acetylation and H3K4 trimethylation (50), mutations in genes encoding the methyltransferase PRDM9 and the histone demethylase KDM5A could contribute to the enhanced inflammatory response upon

minor stimulation with PAMPs, DAMPs, or ACAMPs, or under a state of ER stress in seronegative PR patients, triggering acute flares. Based on results of our whole-exome sequencing, immunophenotyping, and gene expression analyses, we provide a mechanistic model underlying seronegative PR (Figure 6).

Limitations to our multiomic analyses include a lack of samples from the same individual for all scales of analyses. We only had DNA samples from 3 seronegative PR patients; thus, our whole-exome sequencing results could not represent the entire PR patient cohort. In addition, a limitation frequently encountered by large-scale multiomics screening is the inability to reveal detailed information about a particular cell subset (51). We tried to overcome this problem by incorporating FACS and

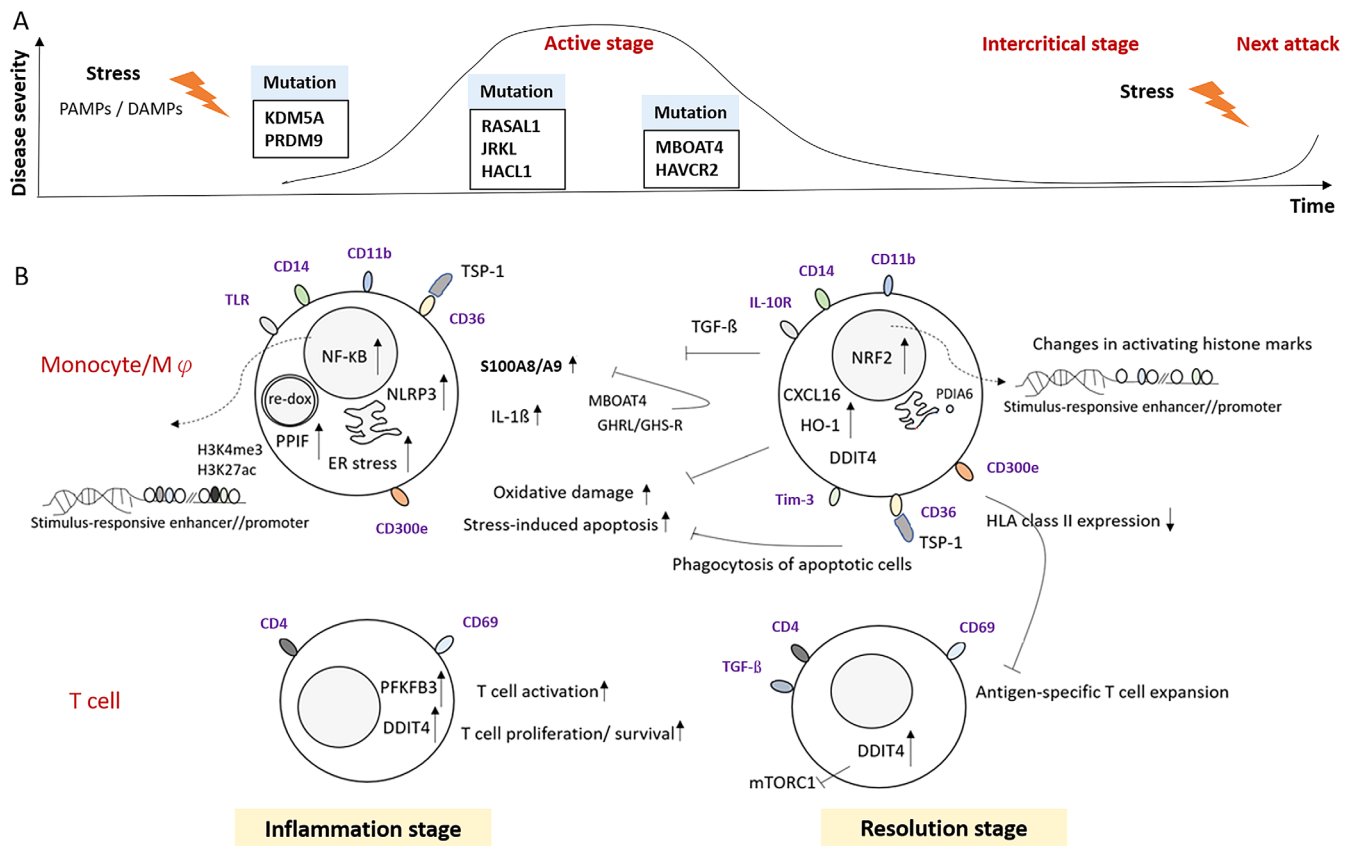


Figure 6. Our model for the pathogenesis of palindromic rheumatism in patients seronegative for rheumatoid factor and anti-cyclic citrullinated peptide antibodies. **A**, Genetic mutations in *KDM5A* and *PRDM9* could contribute to dysregulated stress response and epigenetic remodeling of monocytes, modifying inflammatory response upon restimulation. Genetic mutations in *HAEL1*, *JRKL*, and *RASAL1* might further potentiate stress-induced inflammation and cell death. *MBOAT4* and *HAVCR2* mutations might affect the resolution of inflammation. Mutation in *HAVCR2* might also affect the efficiency of apoptotic cell clearance. **B**, We hypothesized that **under minor stress, such as mild infection, menstrual cycle, or overwork**, PAMPs or DAMPs trigger proinflammatory stress responses in the distinct CD14+CD11b+CD36+ cell subset, secreting inflammatory cytokines, thus contributing to the swelling and pain during an acute flare. Yet, the CD14+CD11b+CD36+ subset could also exert antiinflammatory effects after the phagocytosis of apoptotic cells, possibly via interleukin-10 receptor (IL-10R) and transforming growth factor β (TGF β) expression, or mechanisms related to the induction of regulatory CD4+CD25[−]CD69+ T cells, ultimately leading to the resolution of inflammation. CD4+CD25[−]CD69+ T cells are activated with enriched expression of *DDIT4* and *PFKFB3*, and receive signals from CD14+CD11b+CD36+ cells, limiting antigen-specific T cell expansion and proinflammatory T helper cell differentiation. Decreased *PDI6* expression suggests dysregulated endoplasmic reticulum (ER) stress response. TLR = Toll-like receptor; TSP-1 = thrombospondin 1; GHRL/GHS-R = ghrelin/growth hormone secretagogue receptor; Tim-3 = T cell immunoglobulin and mucin domain-containing protein 3; mTORC1 = mechanistic target of rapamycin complex 1.

subsequent RNA sequencing of the distinct cell subsets into our experiments. However, due to the small amount of CD14+CD11b+CD36+ and CD4+CD25-CD69+ cells in blood, we performed further real-time PCR and functional immune assays using monocytes or lymphocytes instead of the sorted rare cell subsets. Further investigations on the expression signature of downstream molecules potentially involving the dysregulated unfolded protein response process need to be done by specific cell type.

In conclusion, we identified a novel CD14+CD11b+CD36+ phagocyte subset and a distinct CD4+CD25-CD69+ T cell subset, both with immune-activating and immune-regulatory roles, as potential cellular markers for the diagnosis of seronegative PR. Our study provides evidence that dysfunctional ER stress response, inflammatory apoptotic cell clearance, and preserved immune-regulatory function contribute to the relapsing and remitting autoinflammatory features of seronegative PR. Our study also suggests that drugs targeting unfolded protein response pathology might be considered in treating Taiwanese PR patients who are seronegative for RF and anti-CCP antibodies.

ACKNOWLEDGMENTS

We thank the Genomics Center for Clinical and Biotechnological Applications at the National Core Facility for Biopharmaceuticals, Taiwan, for assistance with sequencing, and the Genomics Research Center Mass Core Facility at the Academia Sinica, Taipei, Taiwan for assistance with mass cytometry analyses. We also thank Ms. Guan-Jun Chen for her technical support.

AUTHOR CONTRIBUTIONS

All authors were involved in drafting the article or revising it critically for important intellectual content, and all authors approved the final version to be published. Dr. Lan had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Study conception and design. Yang, Lan.

Acquisition of data. Yang, Lai, Huang, Lin.

Analysis and interpretation of data. Yang, Li.

REFERENCES

- Mankia K, Emery P. What can palindromic rheumatism tell us? [review]. *Best Pract Res Clin Rheumatol* 2017;31:90–8.
- Mankia K, Emery P. Palindromic rheumatism as part of the rheumatoid arthritis continuum [review]. *Nat Rev Rheumatol* 2019;15:687–95.
- Hench PS, Rosenberg EF. Palindromic rheumatism. A “new” oft recurring disease of joints (arthritis, peri-arthritis, para-arthritis) apparently producing no articular residues: report of thirty-four cases (its relationship to “angioneural arthrosis,” “allergic rheumatism” and “rheumatoid arthritis”). *Proc Mayo Clin* 1941;16:808.
- Sanmarti R, Frade-Sosa B, Morlà R, et al. Palindromic rheumatism: just a pre-rheumatoid stage or something else? *Front Med (Lausanne)* 2021;8:657983.
- Chen HH, Lan JL, Hung GD, et al. Association of ultrasonographic findings of synovitis with anti-cyclic citrullinated peptide antibodies and rheumatoid factor in patients with palindromic rheumatism during active episodes. *J Ultrasound Med* 2009;28:1193–9.
- Salvador G, Gomez A, Vinas O, et al. Prevalence and clinical significance of anti-cyclic citrullinated peptide and antikeratin antibodies in palindromic rheumatism. An abortive form of rheumatoid arthritis? *Rheumatology (Oxford)* 2003;42:972–5.
- Guerne PA, Weisman MH. Palindromic rheumatism: part of or apart from the spectrum of rheumatoid arthritis. *Am J Med* 1992;93:451–60.
- Chen HH, Chen DY, Hsieh TY, et al. Predicting the progression of palindromic rheumatism to rheumatoid arthritis: the role of ultrasonography and anti-cyclic citrullinated peptide antibodies. *J Med Ultrasound* 2010;18:17–26.
- Chen HH, Chao WC, Liao TL, et al. Risk of autoimmune rheumatic diseases in patients with palindromic rheumatism: a nationwide, population-based, cohort study. *PLoS One* 2018;13:e0201340.
- Lu LY, Keng HM, Chu JJ, et al. TNF receptor I polymorphism is associated with persistent palindromic rheumatism. *Scand J Rheumatol* 2007;36:278–84.
- Kim SK, Lee HS, Lee KW, et al. Palindromic rheumatism: different genetic background implies a distinct disease entity. *Ann Rheum Dis* 2006;65:1539–40.
- Qiu P, Simonds EF, Bendall SC, et al. Extracting a cellular hierarchy from high-dimensional cytometry data with SPADE. *Nat Biotechnol* 2011;29:886–91.
- Aletaha D, Neogi T, Silman AJ, et al. 2010 rheumatoid arthritis classification criteria: an American College of Rheumatology/European League Against Rheumatism collaborative initiative. *Arthritis Rheum* 2010;62:2569–81.
- Xia C, Braunstein Z, Toomey AC, et al. S100 proteins as an important regulator of macrophage inflammation [review]. *Front Immunol* 2018;8:1908.
- Perera PY, Mayadas TN, Takeuchi O, et al. CD11b/CD18 acts in concert with CD14 and Toll-like receptor (TLR) 4 to elicit full lipopolysaccharide and taxol-inducible gene expression. *J Immunol* 2001;166:574–81.
- Chen B, Miller AL, Rebelatto M, et al. S100A9 induced inflammatory responses are mediated by distinct damage associated molecular patterns (DAMP) receptors in vitro and in vivo. *PLoS One* 2015;10:e0115828.
- Leclerc E, Fritz G, Vetter SW, et al. Binding of S100 proteins to RAGE: an update [review]. *Biochim Biophys Acta* 2009;1793:993–1007.
- Barth ND, Marwick JA, Vendrell M, et al. The “phagocytic synapse” and clearance of apoptotic cells [review]. *Front Immunol* 2017;8:1708.
- Savill J, Dransfield I, Gregory C, et al. A blast from the past: clearance of apoptotic cells regulates immune responses [review]. *Nat Rev Immunol* 2002;2:965–75.
- Baines CP, Kaiser RA, Purcell NH, et al. Loss of cyclophilin D reveals a critical role for mitochondrial permeability transition in cell death. *Nature* 2005;434:658–62.
- Coletta S, Salvi V, Bella CD, et al. The immune receptor CD300e negatively regulates T cell activation by impairing the STAT1-dependent antigen presentation. *Sci Rep* 2020;10:16501.
- Vijayan V, Wagener FADTG, Immenschuh S. The macrophage heme-heme oxygenase-1 system and its role in inflammation. *Biochem Pharmacol* 2018;153:159–67.
- Lepore F, D’Alessandro G, Antonangeli F, et al. CXCL16/CXCR6 axis drives microglia/macrophages phenotype in physiological conditions and plays a crucial role in glioma. *Front Immunol* 2018;9:2750.
- Kim MJ, Sun HJ, Song YS, et al. CXCL16 positively correlated with M2-macrophage infiltration, enhanced angiogenesis, and poor prognosis in thyroid cancer. *Sci Rep* 2019;9:13288.

25. Han Y, Guo Q, Zhang M, et al. CD69⁺ CD4⁺ CD25[−] T cells, a new subset of regulatory T cells, suppress T cell proliferation through membrane-bound TGF- β 1. *J Immunol* 2009;182:111–20.
26. Lee JG, Jaeger KE, Seki Y, et al. Human CD36^{hi} monocytes induce Foxp3⁺CD25⁺ T cells with regulatory functions from CD4 and CD8 subsets. *Immunology* 2021;163:293–309.
27. Whitney ML, Jefferson LS, Kimball SR. ATF4 is necessary and sufficient for ER stress-induced upregulation of REDD1 expression. *Biochem Biophys Res Commun* 2009;379:451–5.
28. Reuschel EL, Wang J, Shivers DK, et al. REDD1 is essential for optimal T cell proliferation and survival. *PLoS One* 2015;10:e0136323.
29. Qin Y, Li B, Arumugam S, et al. m⁶A mRNA methylation-directed myeloid cell activation controls progression of NAFLD and obesity. *Cell Rep* 2021;37:109968.
30. Appenzeller-Herzog C, Hall MN. Bidirectional crosstalk between endoplasmic reticulum stress and mTOR signaling [review]. *Trends Cell Biol* 2012;22:274–82.
31. Zhang Y, Liu W, Wu M, et al. PFKFB3 regulates lipopolysaccharide-induced excessive inflammation and cellular dysfunction in HTR-8/Svneo cells: implications for the role of PFKFB3 in preeclampsia. *Placenta* 2021;106:67–78.
32. Yang Z, Fujii H, Mohan SV, et al. Phosphofructokinase deficiency impairs ATP generation, autophagy, and redox balance in rheumatoid arthritis T cells. *J Exp Med* 2013;210:2119–34.
33. Peng HY, Lucavs J, Ballard D, et al. Metabolic reprogramming and reactive oxygen species in T cell immunity. *Front Immunol* 2021;12:652687.
34. Rico MC, Manns JM, Driban JB, et al. Thrombospondin-1 and transforming growth factor β are pro-inflammatory molecules in rheumatoid arthritis. *Transl Res* 2008;152:95–8.
35. Li A, Song NJ, Riesenberger BP, et al. The emerging roles of endoplasmic reticulum stress in balancing immunity and tolerance in health and diseases: mechanisms and opportunities [review]. *Front Immunol* 2019;10:3154.
36. Hotamisligil GS. Endoplasmic reticulum stress and the inflammatory basis of metabolic disease [review]. *Cell* 2010;140:900–17.
37. Eletto D, Eletto D, Dersh D, et al. Protein disulfide isomerase A6 controls the decay of IRE1 α signaling via disulfide-dependent association. *Mol Cell* 2014;53:562–76.
38. Jarlborg M, Courvoisier DS, Lamacchia C, et al. Serum calprotectin: a promising biomarker in rheumatoid arthritis and axial spondyloarthritis. *Arthritis Res Ther* 2020;22:105.
39. Corradini D, Di Matteo A, Emery P, et al. How should we treat palindromic rheumatism? A systematic literature review. *Semin Arthritis Rheum* 2021;51:266–77.
40. Fujita Y, Matsuoka N, Temmoku J, et al. Hydroxychloroquine inhibits IL-1 β production from amyloid-stimulated human neutrophils. *Arthritis Res Ther* 2019;21:250.
41. Jang CH, Choi JH, Byun MS, et al. Chloroquine inhibits production of TNF- α , IL-1 β and IL-6 from lipopolysaccharide-stimulated human monocytes/macrophages by different modes. *Rheumatology (Oxford)* 2006;45:703–10.
42. Gonneaud A, Gagné JM, Turgeon N, et al. The histone deacetylase Hdac1 regulates inflammatory signalling in intestinal epithelial cells. *J Inflamm (Lond)* 2014;11:43.
43. Nokhbehsaim M, Memmert S, Damanaki A, et al. Effect of interleukin-1 β on ghrelin receptor in periodontal cells. *Clin Oral Investig* 2019;23:113–22.
44. Nakayama M, Akiba H, Takeda K, et al. Tim-3 mediates phagocytosis of apoptotic cells and cross-presentation. *Blood* 2009;113:3821–30.
45. Wolf Y, Anderson AC, Kuchroo VK. TIM3 comes of age as an inhibitory receptor [review]. *Nat Rev Immunol* 2020;20:173–85.
46. Lee J, Phong B, Egloff AM, et al. TIM polymorphisms—genetics and function [review]. *Genes Immun* 2011;12:595–604.
47. Li X, Chen G, Li Y, et al. Involvement of T cell Ig mucin-3 (Tim-3) in the negative regulation of inflammatory bowel disease. *Clin Immunol* 2010;134:169–77.
48. Ortiz-Fernández L, Carmona FD, Montes-Cano MA, et al. Genetic analysis with the immunochip platform in Behcet disease. Identification of residues associated in the HLA class I region and new susceptibility loci. *PLoS One* 2016;11:e0161305.
49. Kaser A, Martínez-Naves E, Blumberg RS. Endoplasmic reticulum stress: implications for inflammatory bowel disease pathogenesis [review]. *Curr Opin Gastroenterol* 2010;26:318–26.
50. Glass CK, Natoli G. Molecular control of activation and priming in macrophages [review]. *Nat Immunol* 2016;17:26–33.
51. Jin H, Goossens P, Juhasz P, et al. Integrative multiomics analysis of human atherosclerosis reveals a serum response factor-driven network associated with intraplaque hemorrhage. *Clin Transl Med* 2021;11:e458.